

Question Paper

Exam Date & Time: 05-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Computer Networks [IT_3127]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Assume a data packet of 1,000 bytes sent over a 20 Mbps wireless link (5)
A) spanning 800 km. The propagation speed in the medium is 2×10^8 m/s. If the processing delay is 1 ms and the average queuing delay is 4 ms, examine the total delay experienced by the packet.
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Apply your understanding of the TCP/IP model to explain how its different layers and associated (3)
protocols work together to achieve end-to-end communication across heterogeneous networks. Illustrate how routing, addressing, and error control are managed within the model.
Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Assume a data packet of size 2000 bytes to be transmitted over a communication link having a (2)
bandwidth capacity of 10 Mbps. Assuming ideal conditions with negligible propagation and queuing delays, determine the transmission delay experienced by the packet on this link.
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) A page has 1 HTML (100 KB) and 5 images (200 KB each), RTT = 50 ms, link rate = 20 Mb/s. (5)
Examine total time to load the page using Non-persistent HTTP.
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Apply your understanding of FTP to explain how it manages file transfers between a local computer (3)
and a remote web server using separate control and data connections. Illustrate how maintaining two connections enhances the efficiency of the file transfer process.
Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Explain how the transport layer manages multiple simultaneous website connections on the same (2)
device. Illustrate your answer with a suitable example showing how different port numbers are used to distinguish each connection.
Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Examine the following dump of UDP header in hexadecimal format: CB84000D001C001C and (5)
compute:
A) I) What is the source port number?

- II) What is the destination port number?
- III) What is the total length of user datagram?
- IV) What is the length of data?
- V) Is the packet directed from a client to server or vice versa?

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Make use of Stop-and-Wait ARQ protocol, the sender transmits Packet 0, but the acknowledgment (ACK1) is lost in transit. Explain what action the sender and receiver will take next, and how the protocol ensures that no duplicate data is delivered to the receiver. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) During a TCP connection setup, a client sends a SYN segment with an initial sequence number of 1200. The server replies with a SYN-ACK having its own sequence number 4500 and acknowledgment number 1201. Explain what the client sends next to complete the handshake, and how these sequence and acknowledgment numbers ensure reliable synchronization between both ends. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) Assume an organization has been assigned a Class C network with a default mask of 255.255.255.0. (5)

- A)
1. The network administrator decides to divide the network into 8 equal subnets. Calculate the new subnet mask and determine the number of hosts per subnet.
 2. Later, the organization decides to combine 8 contiguous Class C networks to create a supernet. Calculate the supernet mask and mention how many total host addresses will be available in this supernet.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Utilize the concept of IPV4 addressing to determine the total number of addresses, the first address, and the last address in a network block that contains the IP address 167.199.170.82/27. Show the steps involved in your calculation. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Explain the concept of Hierarchical Routing in computer networks. Describe how it helps in reducing routing table size and improving scalability in large and complex networks. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Assume the dataword given is 1001, and the generator polynomial is $x^3 + x + 1$. (5)
- A) Find the message transmitted by the sender. If the receiver receives the message, check if the receiver receives the correct message or not.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Assume a sender transmits a dataword 1011, and the codeword generated using even parity is 10111. At the receiver's end, the following codewords are received in different cases: (3)
1. 10111
 2. 10011
 3. 10110

For each case, determine:

- a) Whether an error has occurred or not.
- b) The value of the syndrome.
- c) Whether the original dataword can be correctly recovered.

- C) In a pure ALOHA network, multiple users share a common communication channel. Explain how collisions occur in this system and how slotted ALOHA improves efficiency. (2)

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

IT_3127:Computer Networks

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

IT_3127:Computer Networks

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	4	8(16%)
3	Application	4	12(24%)
4	Analysis	7	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)